

Performance Testing

Date	2 July 2026
Team ID	
Project Name	Rising Waters
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates the Flood Prediction System using Apache JMeter to measure the responsiveness, stability, and reliability of the Flask-based web application and the XGBoost prediction module under simulated user requests.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Apache JMeter
Type of Testing	Load Testing
Target Module / API	Flood Prediction Module (Flask Prediction Endpoint)
Test Environment	Local System (Windows, Python 3.10, Flask)
Test Date	2 July 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Prediction Request Testing	10	5	Prediction requests should be processed successfully
2	Concurrent User Testing	10	5	Application should handle multiple requests without errors
3	Response Time Evaluation	10	5	Response time should remain within acceptable limits
4	Continuous Prediction Requests	10	5	Application should remain stable during repeated requests

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2seconds	17 ms	Pass	Fast prediction response
2	Response Time (Max)	< 5 seconds	59 ms	Pass	Within acceptable limit
3	Throughput (Req/sec)	>20 Req/Sec	11.2 Req/sec	Pass	Successfully handled multiple prediction
4	Error Rate	< 1%	0.00%	Pass	No request failures observed
5	CPU Utilization	< 80%	N/A	N/A	Not measured using Apache JMeter
6	Memory Utilization	< 80%	N/A	N/A	Not measured using Apache JMeter

Step 4: Observations & Analysis

Key Findings

- The Flask web application successfully processed all prediction requests without errors.
- The average response time was **17 ms**, indicating fast prediction performance.
- The application achieved a throughput of **11.2 requests per second** during testing.
- No request failures were observed, demonstrating stable application performance.

Bottlenecks Identified

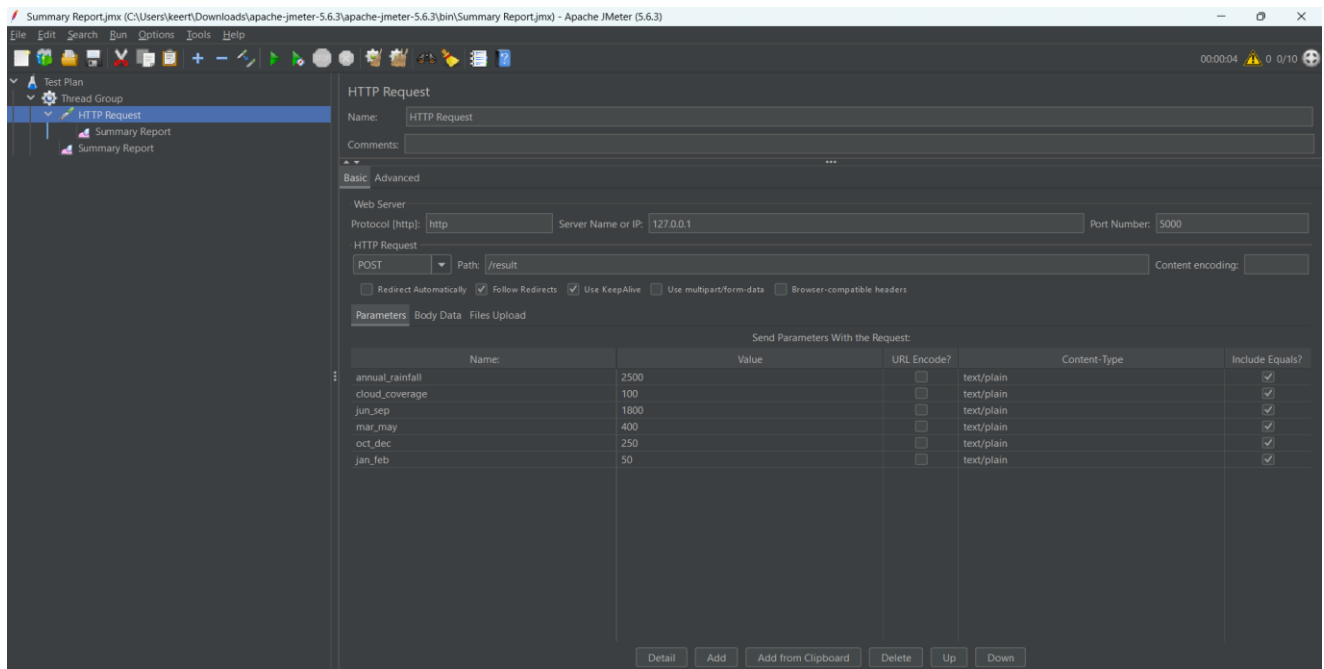
- No significant performance bottlenecks were observed during testing.
- Performance may vary with larger workloads or different deployment environments.

Optimization Steps Taken

- Optimized the XGBoost model integration for efficient prediction.
- Improved data preprocessing before prediction.
- Configured the Flask application to efficiently handle prediction requests.

Step 5: Screenshots / Evidence

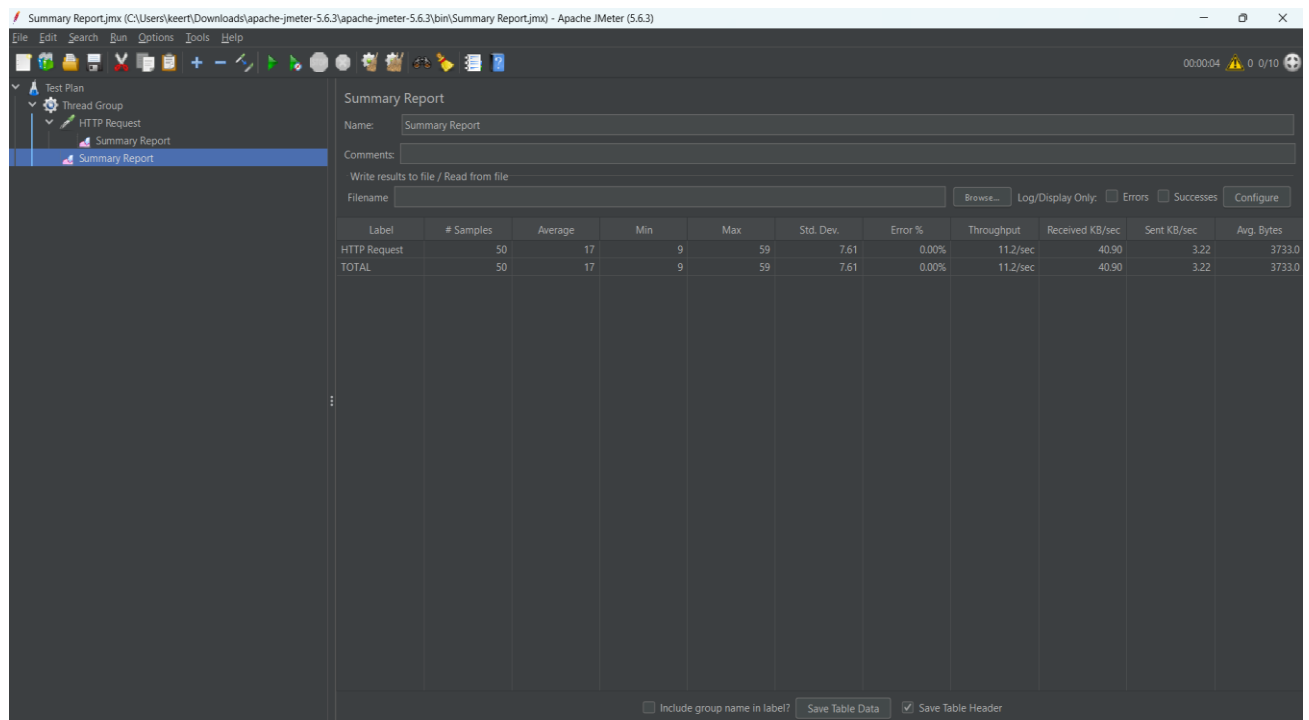
The following screenshots show the performance testing results of the Flood Prediction System using Apache JMeter. They include the Summary Report and View Results Tree, demonstrating the application's response time, throughput, error rate, and successful execution of prediction requests.



The screenshot shows the Apache JMeter interface with the 'HTTP Request' element selected in the Test Plan. The configuration is as follows:

- Name:** HTTP Request
- Comments:**
- Basic Tab:**
 - Web Server:** Protocol (http), Server Name or IP: 127.0.0.1, Port Number: 5000
 - HTTP Request:** Method (POST), Path: /result, Content encoding:
 - Options:** ☐ Redirect Automatically, ☒ Follow Redirects, ☒ Use KeepAlive, ☐ Use multipart/form-data, ☐ Browser-compatible headers
- Parameters Tab:**
 - Send Parameters With the Request:**

Name	Value	URL Encode?	Content-Type	Include Equals?
annual_rainfall	2500	☐	text/plain	☒
cloud_coverage	100	☐	text/plain	☒
jun_sep	1800	☐	text/plain	☒
mar_may	400	☐	text/plain	☒
oct_dec	250	☐	text/plain	☒
jan_feb	50	☐	text/plain	☒

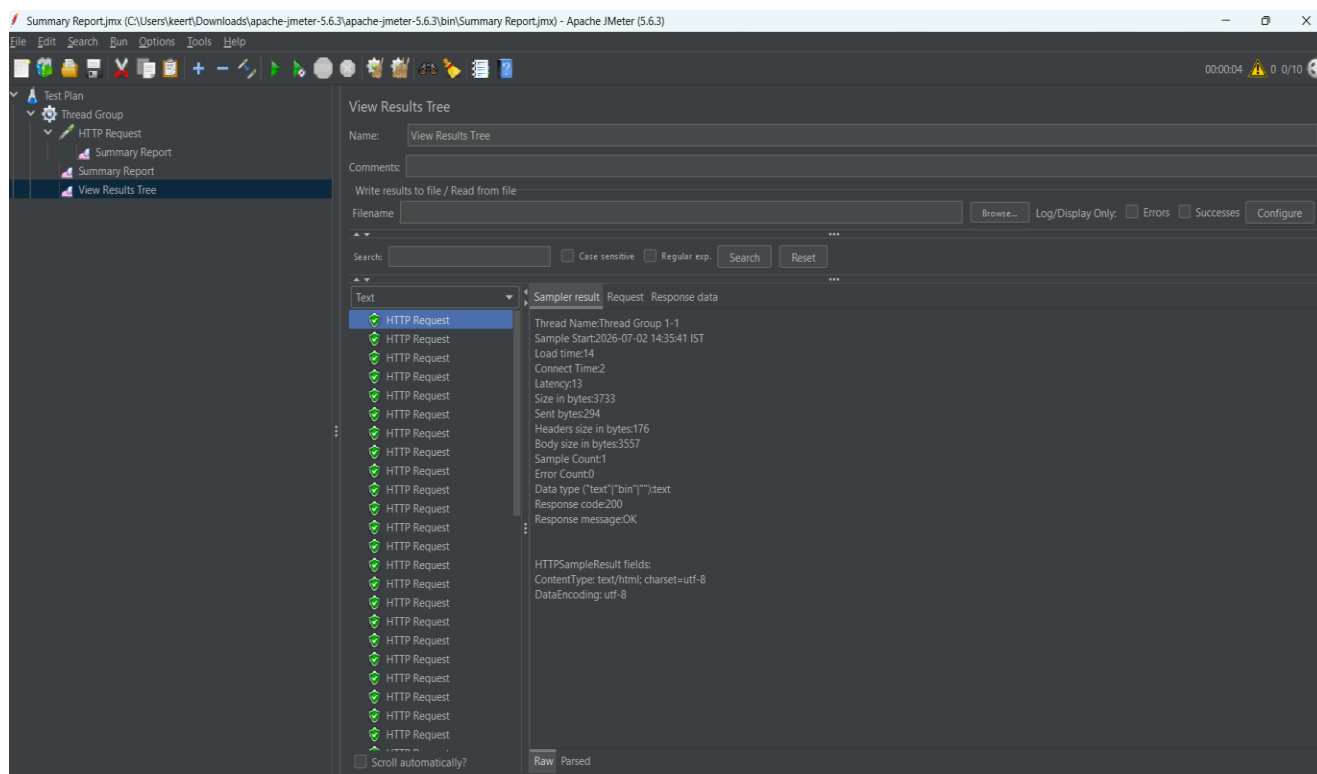


The screenshot shows the Apache JMeter Summary Report for the 'Summary Report' element. The report includes the following data:

Label	# Samples	Average	Min	Max	Std. Dev.	Error %	Throughput	Received KB/sec	Sent KB/sec	Avg. Bytes
HTTP Request	50	17	9	59	7.61	0.00%	11.2/sec	40.90	3.22	3733.0
TOTAL	50	17	9	59	7.61	0.00%	11.2/sec	40.90	3.22	3733.0

Additional options at the bottom include:

- ☐ Include group name in label?
-
- ☒ Save Table Header



Summary Report.jmx (C:\Users\keert\Downloads\apache-jmeter-5.6.3\apache-jmeter-5.6.3\bin\Summary Report.jmx) - Apache JMeter (5.6.3)

File Edit Search Run Options Tools Help

00:00:04 0 0/10

Test Plan

- Thread Group
 - HTTP Request
 - Summary Report
 - Summary Report
 - View Results Tree

View Results Tree

Name: View Results Tree

Comments:

Write results to file / Read from file

Filename: Browse... Log/Display Only: ☐ Errors ☐ Successes

Search: ☐ Case sensitive ☐ Regular exp.

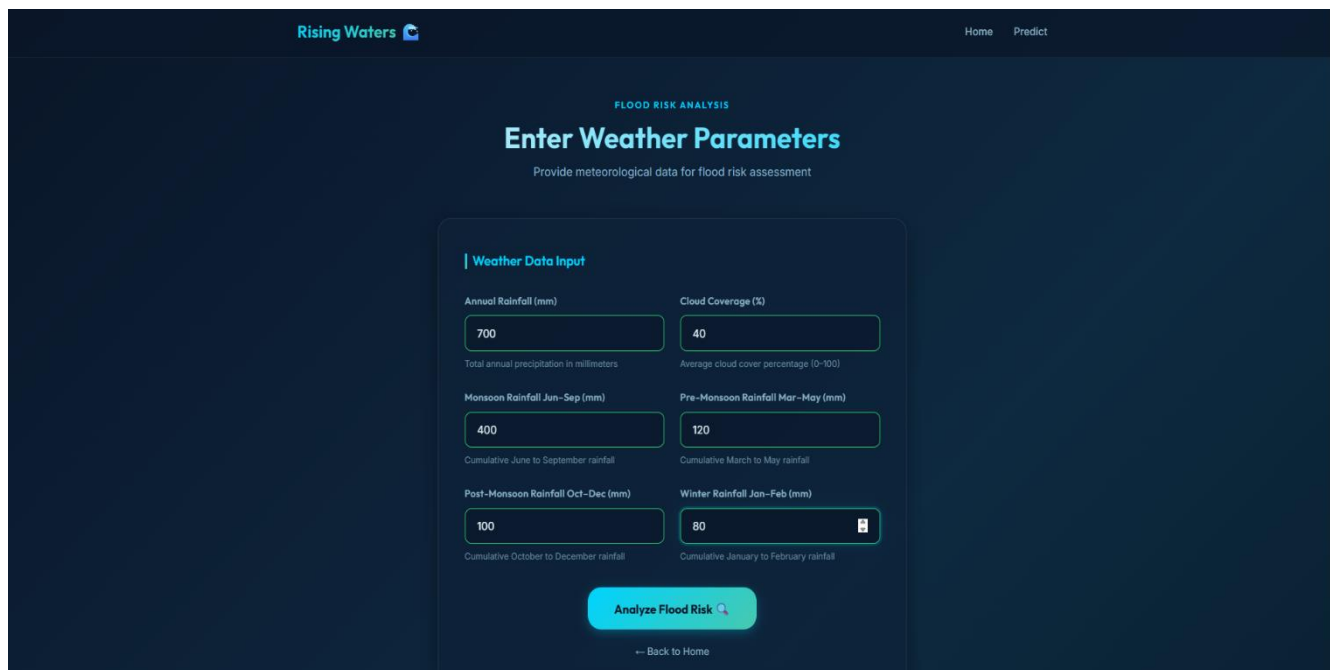
Text

Sampler result: Request Response data

Thread Name: Thread Group 1-1
Sample Start: 2026-07-02 14:35:41 IST
Load time: 14
Connect Time: 2
Latency: 13
Size in bytes: 3733
Sent bytes: 294
Headers size in bytes: 176
Body size in bytes: 3557
Sample Count: 1
Error Count: 0
Data type: "text"["bin"]text
Response code: 200
Response message: OK

HTTPSampleResult fields:
ContentType: text/html; charset=utf-8
DataEncoding: utf-8

☐ Scroll automatically?



Rising Waters  Home Predict

FLOOD RISK ANALYSIS

Enter Weather Parameters

Provide meteorological data for flood risk assessment

Weather Data Input

Annual Rainfall (mm) 700 Total annual precipitation in millimeters	Cloud Coverage (%) 40 Average cloud cover percentage (0-100)
Monsoon Rainfall Jun-Sep (mm) 400 Cumulative June to September rainfall	Pre-Monsoon Rainfall Mar-May (mm) 120 Cumulative March to May rainfall
Post-Monsoon Rainfall Oct-Dec (mm) 100 Cumulative October to December rainfall	Winter Rainfall Jan-Feb (mm) 80 Cumulative January to February rainfall



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